Class 8: Breast Cancer Mini Project

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Background

Today we will employ all R techniques for data analysis we used so far including machine learning methods of clustering and PCA to analyze an actual breast cancer biopsy data.

Data Import

The data is in CSV format:

```
wiscancer <- read.csv("WisconsinCancer.csv", row.names=1)

head(wiscancer,3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
```

```
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

Q1 How many observations are in this dataset?

```r
nrow(wiscancer)
```

```
[1] 569
```

Q2 How many of the observations have a malignant diagnoss?

```r
sum(wiscancer$diagnosis == "M")
```

```
[1] 212
```

```r
table(wiscancer$diagnosis)
```

```
  B   M 
357 212 
```

Q3 How many variables/ features in the data are suffixed with _mean

```r
colnames(wiscancer)
```

```
 [1] "diagnosis"               "radius_mean"
 [3] "texture_mean"            "perimeter_mean"
 [5] "area_mean"               "smoothness_mean"
 [7] "compactness_mean"        "concavity_mean"
 [9] "concave.points_mean"     "symmetry_mean"
[11] "fractal_dimension_mean"  "radius_se"
[13] "texture_se"              "perimeter_se"
[15] "area_se"                 "smoothness_se"
[17] "compactness_se"          "concavity_se"
[19] "concave.points_se"       "symmetry_se"
[21] "fractal_dimension_se"    "radius_worst"
[23] "texture_worst"           "perimeter_worst"
[25] "area_worst"              "smoothness_worst"
[27] "compactness_worst"       "concavity_worst"
[29] "concave.points_worst"    "symmetry_worst"
[31] "fractal_dimension_worst"
```

```r
length(grep("_mean",colnames(wiscancer)))
```

```
[1] 10
```

We need to remove the `diagnosis` column before doing any more analysis of this dataset since we don't want to pass this to PCA etc. Save it as a separate vector that we can later use to compare findings to those of experts.

```r
#saving diagnosis
diagnosis <- wiscancer [,1]
#dataframe omitting first column
wisc <- wiscancer[,-1]
```

Principal Component Analysis (PCA)

The primary function in base R for performing PCA is `prcomp()`. We use the optional argument `scale = TRUE` because the variables in the dataset are measured on different scales.

```r
wisc.pr <- prcomp(wisc,scale=T)
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

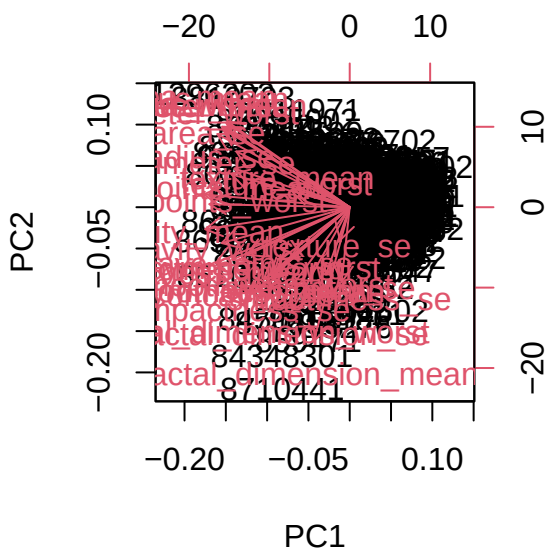
3

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7

Interpreting PCA Plots

```
biplot(wisc.pr)
```

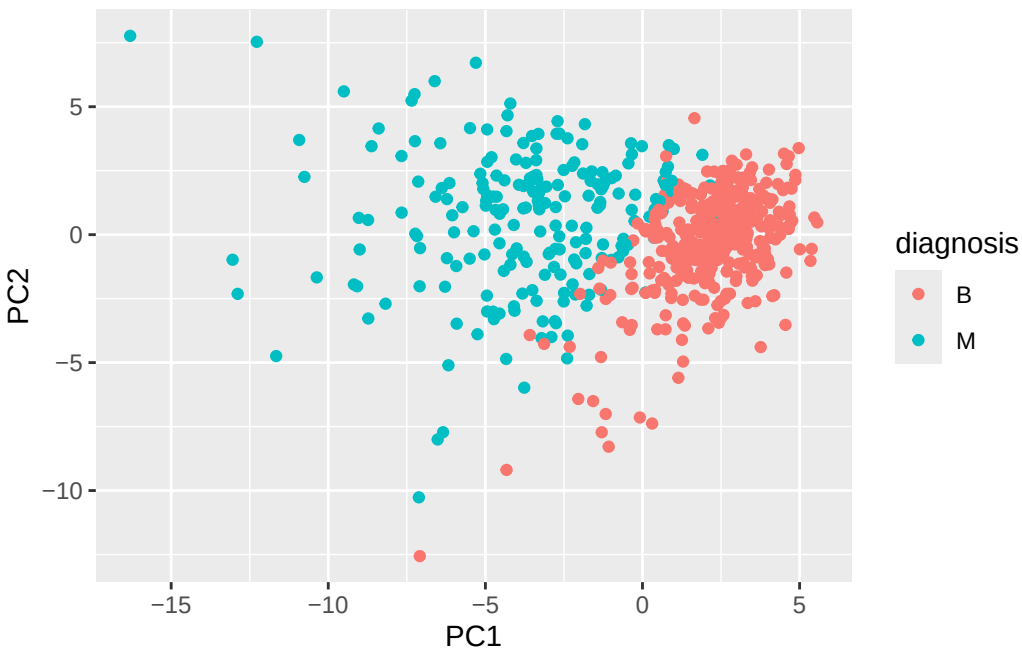


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

It's not a good plot, since not much information is discernable. It is hard to understand due to there just being one large clump of data in the middle, making it too crowded with insufficient scaling to observe correlations visually.

Scatter plot observations by components 1 and 2

```
library(ggplot2)
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```




```
summary(wisc.pr)
```

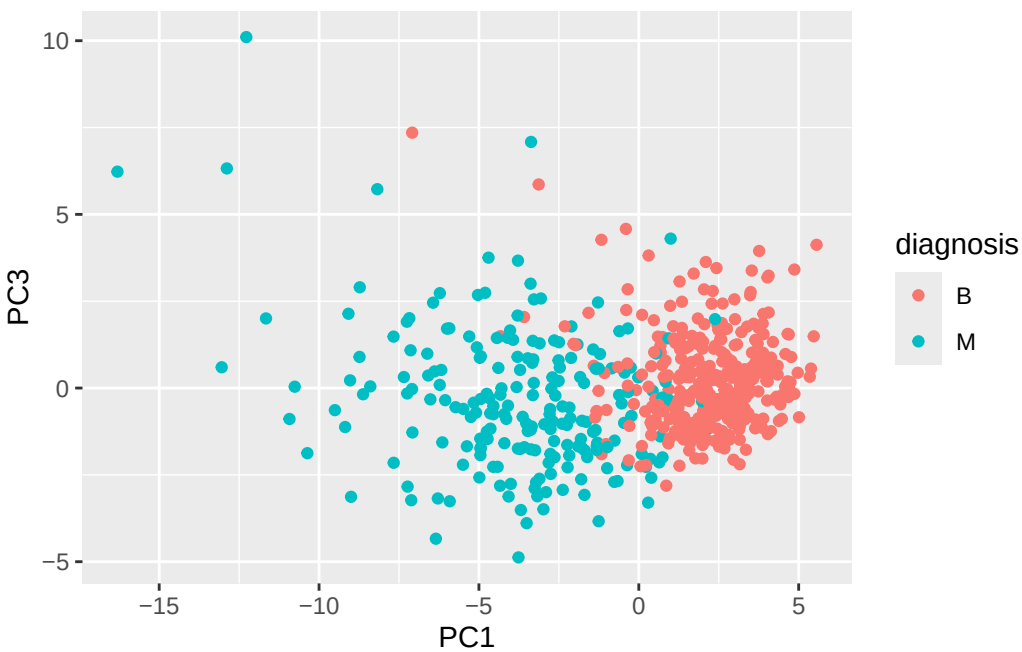
```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
```

```
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The PC1 vs PC2 plot shows clearer separation between malignant and benign samples than PC1 vs PC3

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



Calculate variance of components.

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

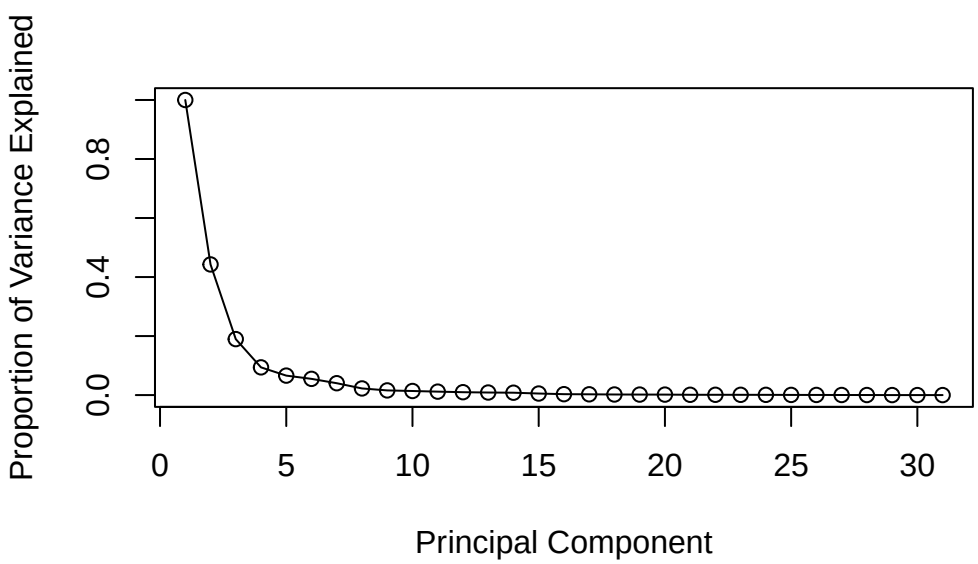
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

Variance explained by each principal component: pve

```
pve <- pr.var/ sum(pr.var)
```

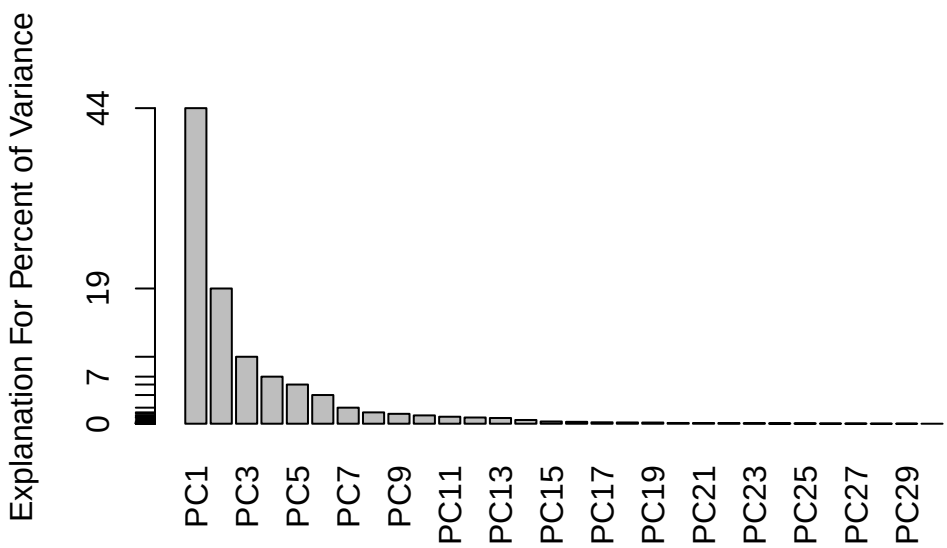
Plot variance explained for each principal component

```
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Alternative scree plot of same data, data driven y-axis

```
barplot(pve, ylab = "Explanation For Percent of Variance",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



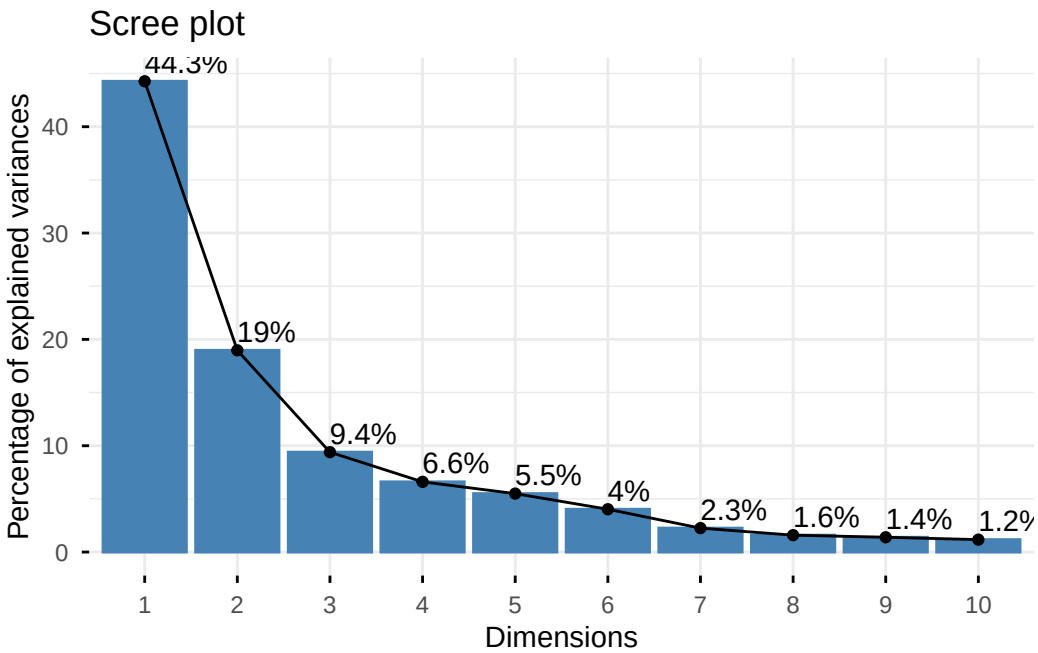
ggplot based graph

```
library(factoextra)
```

```
Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
fviz_eig(wisc.pr, addlabels = TRUE)
```

```
Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
Ignoring empty aesthetic: `width`.
```



Communicating PCA Results

> Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
pc1 <- wisc.pr$rotation [, "PC1"]
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

```
min(pc1)
```

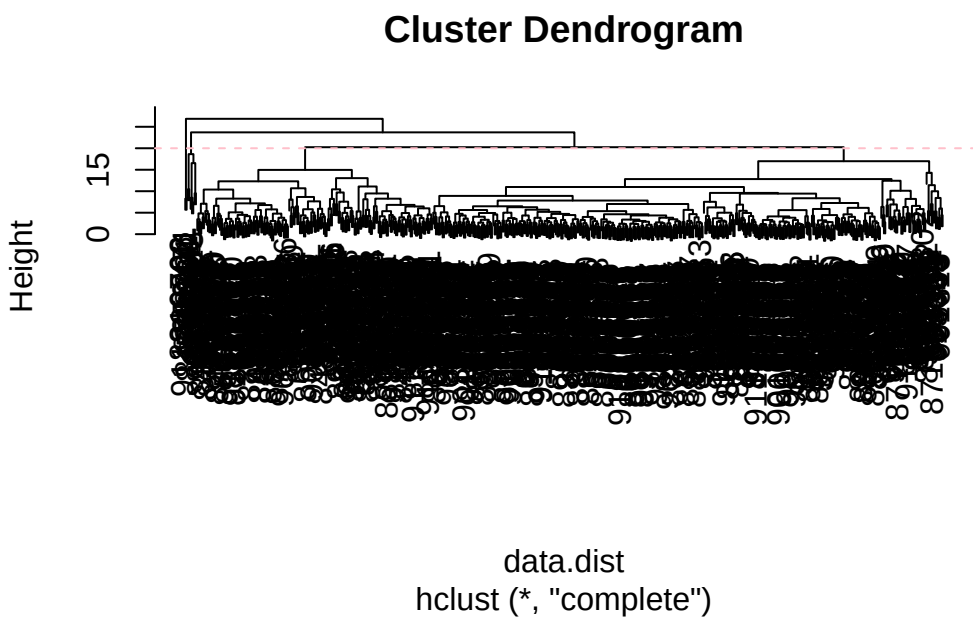
```
[1] -0.2608538
```

The component of the loading vector for `concave.points_mean` on PC1 is -0.261. This feature has one of the largest absolute loadings on PC1, indicating that it is a major contributor to this principal component.

Hierarchial clustering

Scale the wisc.data data using the "scale()" function

```
wisc.data <- wisc
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist)
plot(wisc.hclust)
abline(h=20, col="pink", lty=2)
```



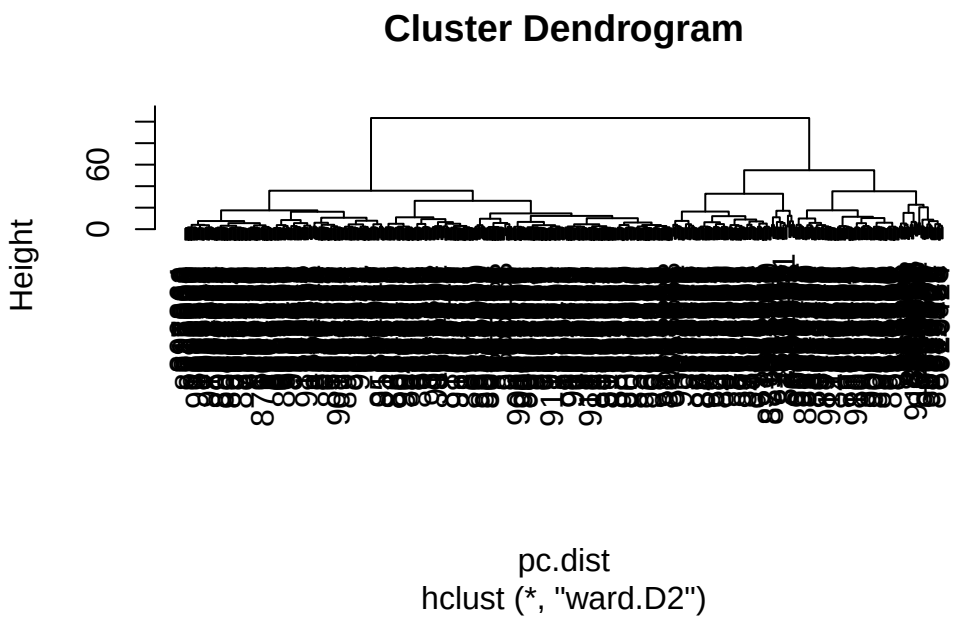
Q10. Using plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

~20

Combining methods

The idea is to use the new variables like the principal component scores stored in `wisc.pr$x` to provide a more concise and informative representation of the dataset than the original features, which were the 30 variables in `wisc.data`, as the basis for clustering analysis.

```r
pc.dist <- dist(wisc.pr$x[, 1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```



Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

I prefer the ward d2 method, as it looks distributed the best. It is more even, while the single method looks too convoluted, the average method has weird clumps, and the complete method looks way too spread apart.

```r
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

```r
table(grps, diagnosis)
```

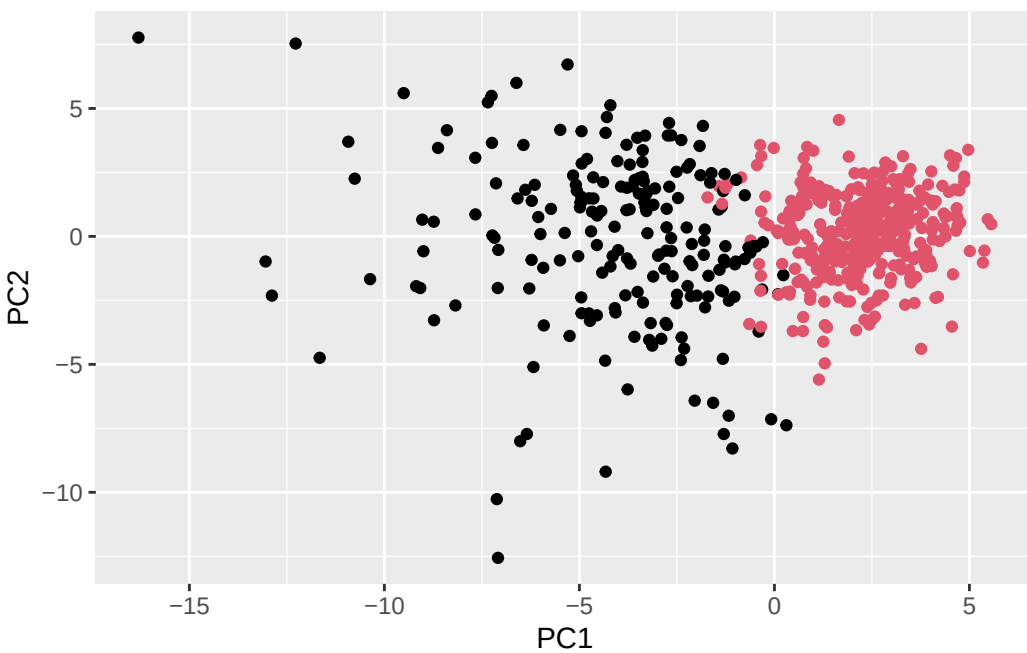
```
    diagnosis
```

```
grps   B   M
   1  24 179
   2 333  33
```

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



Now run `table()` with both clustering `grps` and the expert `diagnosis`

```r
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

Cluster "1" has 179 "M" Diagnosis Cluster "2" has 333 "B" Diagnosis

179 TP 24 FP

333 TN 33 FN

Sensitivity: TP / (TP+FN)

```r
179/(179+33)
```

```
[1] 0.8443396
```

Specificity: TN / (TN+FP)

```r
333/(333+24)
```

```
[1] 0.9327731
```

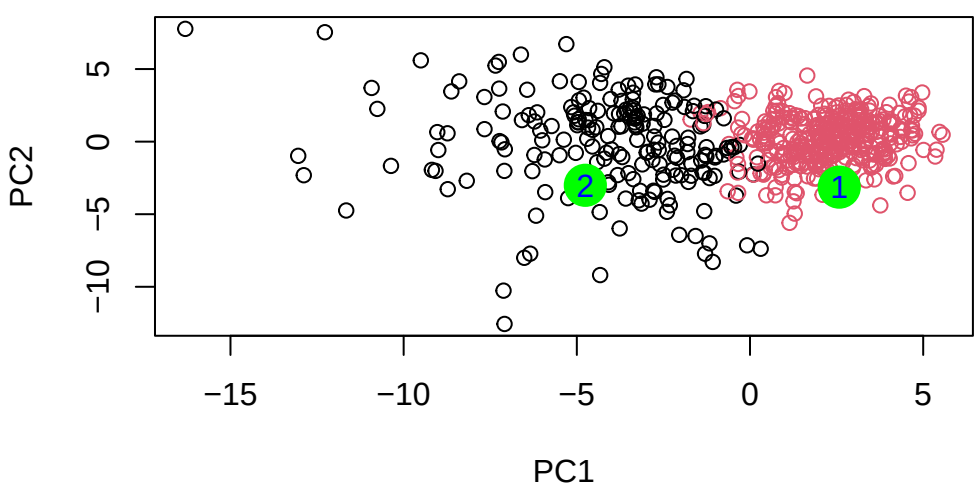
Prediction

We can use our PCA model for prediction of

```r
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="green", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="blue")
```



Q16 Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized for a follow up, as their projected PCA scores place them closer to the malignant cluster.